

OTITIS MEDIA

Neutrophil-mediated delivery of antibiotics across the tympanic membrane for otitis media treatment in a preclinical model

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Acute otitis media (AOM) is a leading cause of pediatric antibiotic prescriptions. Systemic antibiotics cause side effects and antibiotic resistance, whereas local delivery of antibiotics (directly to the middle ear) is hindered by an impermeable biological barrier, the tympanic membrane (TM). Here, we developed hydroxylated liposomes that are internalized by neutrophils recruited during AOM to transport encapsulated ciprofloxacin across the intact TM. In mice, complement opsonization of the hydroxylated liposomes increased neutrophil uptake *ex vivo* and *in vivo*. In chinchillas with AOM induced by nontypeable *Haemophilus influenzae*, a hydrogel formulation containing the hydroxylated liposomes applied topically to the TM led to middle ear concentrations of ciprofloxacin that were 36-fold higher than those achieved by liposome-free hydrogels. The pathogens were undetectable within 24 hours, which compares favorably to the current 5- to 10-day treatment using oral antibiotics. The treated TM appeared identical to the pristine ones, and auditory brainstem response thresholds in healthy animals were unchanged by the treatment. Depleting complement proteins using cobra venom factor reduced the transtympanic delivery efficiency and led to a cure rate of 25%, implicating the essential role of complement opsonization in enabling neutrophil hitchhiking. Unlike previous immune cell-based delivery approaches, our method achieves neutrophil hitchhiking through a simple topical application of hydroxylated liposomes, thereby eliminating the need for invasive neutrophil harvesting. It points to a potential low-cost and noninvasive treatment for this prevalent disease, poised to reduce pediatric antibiotic usage.

INTRODUCTION

Acute otitis media (AOM) is the primary reason for pediatric antibiotic usage. An estimated 19.5 million AOM cases account for 24% of the antibiotic prescriptions written to children in the United States (1) and an annual cost of \$3 billion to \$4 billion (2). Globally, about 80% of children experience at least one episode of AOM by school age (3). AOM is predominantly caused by bacterial infections (4, 5), among which nontypeable *Haemophilus influenzae* (NTHi) accounts for 16 to 61% of all AOM episodes globally (6, 7). The current mainstay treatment for AOM comprises oral antibiotics for 5 to 10 days (8); however, oral antibiotics can cause acute side effects in children, including diarrhea and vomiting (9), and are correlated with long-term health issues like asthma and obesity (10). As a result, the medication is discontinued prematurely in one-third of US children who are 6 to 36 months of age, frequently upon resolution of symptoms (11). This poor patient compliance, in turn, speeds up the development of antibiotic resistance. A treatment strategy to minimize antibiotic usage during AOM treatment could mitigate the side effects and development of drug resistance.

One strategy to reduce antibiotic usage is local delivery to the affected tissue. However, the tympanic membrane (TM), although only around 100 μm in thickness in humans (12), has been a long-standing bottleneck in local drug delivery in the ear because it is largely impermeable to most molecules, including antibiotics. Current local delivery practice often invokes surgical disruption of the

TM (during the placement of a tympanostomy tube). To address the broader population that does not require surgical intervention, non-invasive delivery strategies have been explored extensively (13–18). The current best-in-class formulation uses chemical permeation enhancers (CPEs) to disrupt the structural integrity of the most impermeable layer in the TM, the stratum corneum (SC) connected with the outer ear canal. The CPE-containing formulations have enabled transtympanic antibiotic delivery (14, 15, 19) but are akin to physical disruption, albeit at the molecular level, hence associated with low delivery efficiency and high irritation potential (16).

A recent work has explored the use of immune cells as carriers of drug-loaded cargo across challenging biological barriers such as the blood-brain barrier (20). During AOM, neutrophils, the most abundant leukocytes in the bloodstream (21), undergo chemotaxis, migrating from the bloodstream toward the middle ear (22). However, existing immune cell-mediated therapies often rely on the invasive and costly procedure of harvesting immune cells, loading drugs *ex vivo*, and reintroducing the immune cells back into systemic circulation (20, 23, 24), rendering them unrealistic for AOM treatment. Although neutrophils have been observed in skin during psoriasis and dermatophytosis (25, 26), their ability to cross the SC layer has not been reported. Therefore, the possibility of using neutrophils to overcome the SC barrier and deliver antibiotics locally remains unexplored.

In this study, motivated by our observation that neutrophils (around 12 μm in diameter) are present throughout the TM during AOM, we hypothesized that neutrophils can be leveraged as carriers to overcome the impermeable TM. Specifically, this design harnesses neutrophils as endogenous micromotors to traffic the antibiotic cargo that is placed in the patient's outer ear canal across the TM, targeting

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the site of infection. We predicted that their active chemotaxis to the infected middle ear could enable highly efficient and biocompatible topical treatment while avoiding systemic antibiotic exposure. To use innate neutrophils, we devised a neutrophil-targeting liposome strategy and used ciprofloxacin (Cip) as an exemplar AOM therapeutic. The liposome design centers around the complement cascade, a key component of the innate immunity and host defense system (27). Neutrophils responding to infections such as AOM internalize foreign substances that have been opsonized by complement molecules (28), mostly by the cleavage products of the complement 3 (C3) protein, that is, C3b and its subsequent cleavage molecule, iC3b (29, 30). Prior works demonstrated that opsonization on foreign substances could be engineered by a reaction between a short-lived thioester moiety in C3b and hydroxyl (OH) groups (27, 31). As such, we proposed a delivery system containing OH-bearing liposomes made of 1,2-distearoyl-*sn*-glycero-3-phosphoglycerol (DSPG). The C3b fragments covalently linked to the liposome serve as targets for enhanced internalization and trafficking by innate neutrophils to the infected middle ear.

Current AOM treatments commonly use the antibiotic Cip where it can be applied to the middle ear through a perforated TM (32, 33). Although amoxicillin is chosen over Cip for the mainstay standard of care, a multiday course of oral antibiotics (34, 35), because of the harmful effects of Cip on developing joints when administered systemically (36, 37), Cip is commonly used in ear drops such as Ciprodex (38, 39). Cip has also been encapsulated in a thermoresponsive hydrogel made of Poloxamer 407 (P407) to achieve sustained delivery, as demonstrated in a US Food and Drug Administration (FDA)-approved otic suspension product, Otiprio, which treats patients with OM undergoing tympanostomy. Clinical trial results have indicated high biocompatibility with no formulation-related audiometric change (40, 41). Despite the effectiveness and safety of Cip in treating AOM, its permeation across the intact TM, even enhanced by CPEs, has achieved limited efficiency. The inclusion of CPEs further introduces formulation challenges, such as drug-excipient complexation-induced precipitation (16).

To overcome these challenges while building on the efficacy of Cip, we encapsulated it in DSPG liposomes, which were subsequently loaded into a P407 hydrogel. The formulation is a solution under room temperature for ease of administration into the outer ear canal and gels firmly upon reaching the TM. As such, a single dose (for example, applied in the pediatrician's office) could sustain the release of antibiotics over the 7-day treatment, avoiding the premature discontinuation of the treatment. The neutrophil-based delivery enables the eradication of AOM using a small amount of antibiotics, that is, 0.5 mg of Cip, corresponding to 6.25% of that in the best-in-class transtympanic formulation (19) and 0.5% of that used in systemic AOM treatment (42). The local administration and high transtympanic delivery efficiency combined minimize antibiotic exposure during AOM treatment.

To address some of the central challenges of transtympanic drug delivery, including antibiotic dose and compliance, we tested a strategy leveraging the complement system to harness innate neutrophils as a drug delivery vehicle. This strategy enables targeted delivery without the requirement for specific targeting biomolecules, such as antibodies, that are costly to deploy for such a prevalent disease. The transtympanic delivery platform has the potential to reduce pediatric antibiotic usage globally, thus mitigating acute and long-term health issues related to antibiotic exposure early in life.

RESULTS

Hydroxyl-bearing liposomes traverse the TM of infected animals by hitchhiking on innate neutrophils through the opsonization by complement molecules

To first investigate the trafficking of neutrophils during AOM, infection was induced by direct inoculation of NTHi into the auditory bullae of *Chinchilla lanigera*, an established preclinical model for the pathophysiology and treatment of AOM (43, 44). Although inflammation or neutrophils were absent from the TM under normal homeostasis (fig. S1A), neutrophils were observed in the TM and auditory bullae, excised at 48 or 72 hours postinfection (fig. S1, B and C). There are around 30 neutrophils visible in the histology image shown, permeating the entire TM structure. Neutrophils were also captured in the middle ear fluid (MEF) (fig. S1, D and E), which is consistent with prior reports of 40 to 300 neutrophil counts in the middle ear cavity after an NTHi inoculation (45, 46). On the basis of a quantitative analysis of histological images from a 280 μm by 370 μm area, the neutrophil counts at 48 and 72 hours bear no significant difference ($P > 0.05$) (fig. S1F). The identification of neutrophils in the TM during AOM informed the design of the liposomal formulation, as detailed below.

We hypothesized that liposomes bearing hydroxyl moieties would experience enhanced opsonization by C3b fragments, which marks the liposome for internalization as a way to “load” the liposomes into neutrophils for transtympanic delivery (fig. S2). To verify the presence of the C3b fragments in chinchillas with AOM, we analyzed their sera and MEFs using Western blotting (fig. S3, A to B). Note that an anti-guinea pig complement antibody was used in the Western blot because of an incomplete protein database on *C. lanigera*; similar methods have been used in the literature (47–49). The Western blot identified the C3 α (120 kDa) and iC3b α' fragments (67, 46, and 40 kDa) in both fluids. The molecular weight of the C3 fragments was consistent with our prediction based on the known C3 cleavage mechanism by the alternative activation pathway (fig. S3C) (47, 50).

To assess the effectiveness of the neutrophil internalization, we compared hydroxyl-bearing liposomes made of DSPG and non-hydroxyl-bearing ones made of dioleoyl-3-trimethylammonium propane (DOTAP) (Fig. 1A). DOTAP was chosen because it is positively charged and thus considered effective for penetrating the negatively charged SC, the most impermeable structure in the TM (51–53). The liposomes were made in a stealth manner by incorporating polyethylene glycol (PEG)-conjugated lipid (1,2-distearoyl-*sn*-glycero-3-phosphoethanolamine-PEG) (at a 5% molar ratio) in the liposome to form a hydrated surface layer, which is known to reduce nonspecific interactions with the reticuloendothelial system (54). As such, the liposomes are denoted as sLip, standing for stealth liposomes. Dynamic light scattering (DLS) measured an average diameter of 127.4 ± 5.2 nm and an average surface charge of -20.1 ± 0.4 mV for the DSPG liposomes (DSPG/sLip) and 134.2 ± 1.5 nm and $+26.8 \pm 1.5$ mV, respectively, for the DOTAP liposomes (DOTAP/sLip) (fig. S4 and table S1).

The enhanced opsonization of the hydroxyl-bearing liposomes was illustrated by analyzing their protein coronas using Western blotting (55–58). The protein coronas were formed by incubating the liposomes with the MEFs harvested from infected chinchillas. C3b fragments strongly bind to DSPG/sLip but not to DOTAP/sLip (Fig. 1B, top). The amounts of the complement biomarkers displayed on the surfaces of DSPG/sLip were 6.6-fold higher than those on DOTAP/sLip ($P = 0.0005$). Similar observations were made when

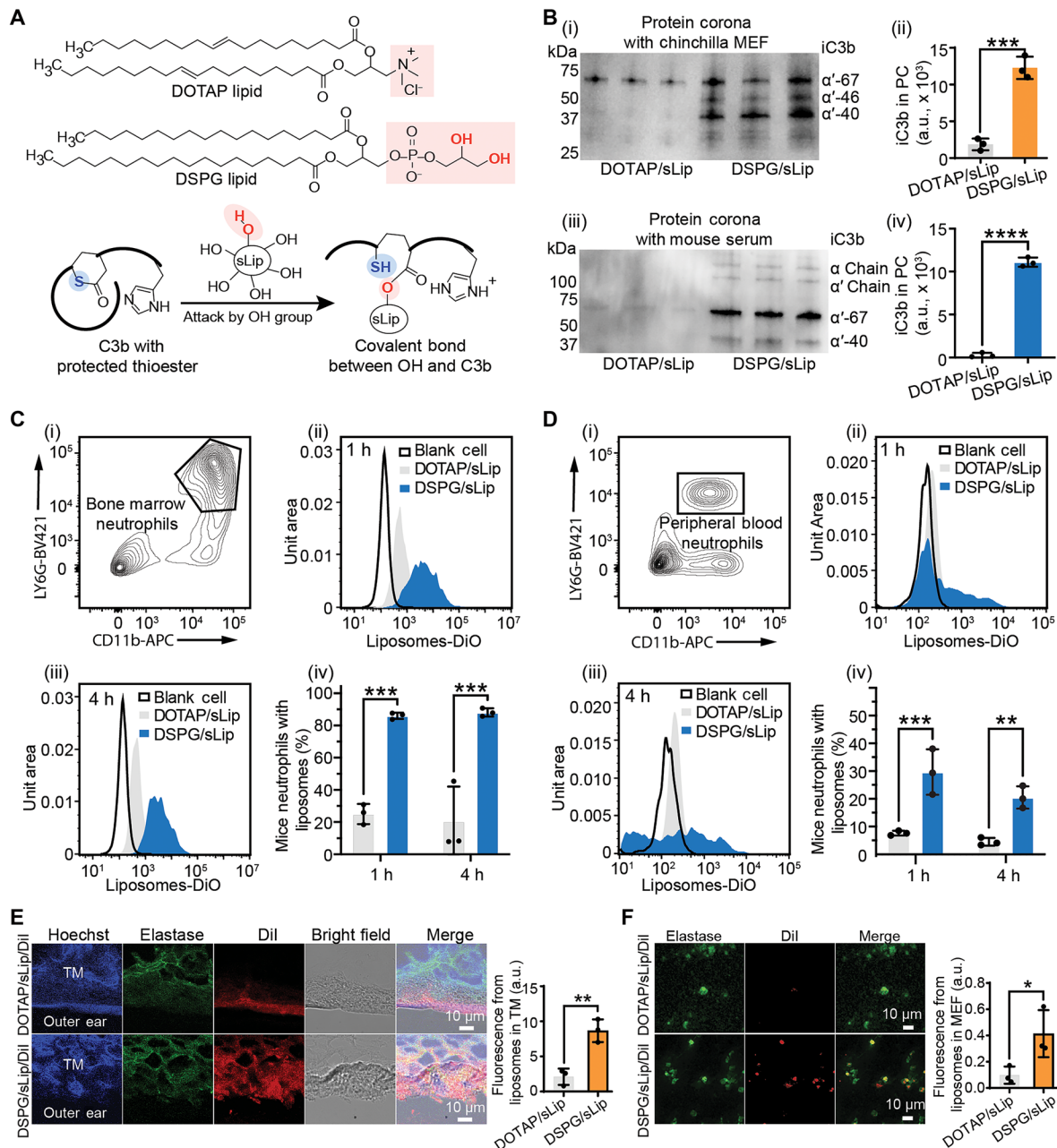


Fig. 1. Hydroxyl-bearing liposomes traverse the TM through neutrophil uptake, enabled by a reaction with the complement protein fragment C3b. (A) Molecular structures of a non-hydroxyl-bearing lipid, DOTAP, and a hydroxyl-bearing lipid, DSPG, and the reaction between the DSPG liposome hydroxyl groups (DSPG/sLip) and C3b that marks DSPG/sLip for subsequent neutrophil internalization. (B) Western blot analysis and quantification of iC3b fragments in the DOTAP/sLip and DSPG/sLip protein coronas after incubating with the MEFs collected from chinchillas with AOM (i and ii) ($***P = 0.0005$) and BALB/c mice sera (iii and iv) ($****P < 0.0001$). a.u., arbitrary units. (C) Flow cytometry gating of neutrophils from BALB/c bone marrow by CD11b-APC. (i). Histogram of ex vivo uptake of the opsonized liposomes by neutrophils ex vivo for 1 hour (ii) and 4 hours (iii); the liposomes were fluorescently labeled using DiO. Quantification of the fraction of liposome-containing neutrophils (iv) ($***P < 0.001$). (D) In vivo uptake by peripheral blood neutrophils (i) after intravenous injection (50 mg of lipid per kg body weight) of the two types of liposomes in BALB/c mice at 1 hour (ii) and 4 hours (iii). Quantification of the fraction of liposome-containing neutrophils (iv) ($**P = 0.0058$ and $***P = 0.0008$). (E) Fluorescence images of the auditory bullae of chinchillas with AOM treated with the two liposomal formulations and the amounts of each liposome in the TM ($**P = 0.0047$), estimated from their fluorescence intensities; liposomes are fluorescently labeled using Dil (red); DSPG/sLip/Dil readily traverse the TM (blue, with Hoechst staining for cell nucleus) by internalization by the native neutrophils (green, with elastase staining), whereas DOTAP/sLip/Dil remain largely in the SC layer. (F) Fluorescence images of the MEFs of chinchillas with AOM treated with the two liposomal formulations and the amounts of each liposome in the MEFs ($*P = 0.0454$), estimated from their fluorescence intensities. For all panels, $n = 3$ for each group; data are presented as means $\pm$ SD, and dots represent individual participants; scale bars, 10 μ m; P values were determined by two-tailed unpaired Student's t tests in (B), (E), and (F) and one-way ANOVA with Tukey's test in (C) and (D); $*P < 0.05$, $**P < 0.01$, $***P < 0.001$, and $****P < 0.0001$. See fig. S6 for the gating strategy of neutrophil subsets in (C) and (D); see fig. S7 for additional fluorescence micrographs of the group treated with DSPG/sLip/Dil and fig. S8 for the low-magnification micrographs for both groups in (E).

the liposomes were incubated with the sera of chinchillas with AOM (fig. S5). The C3b fragments were enriched in the protein coronas of DSPG/sLip ($P = 0.0082$ versus DOTAP/sLip) given that the complement cascade is a systemic reaction.

To provide direct evidence that the hydroxylated liposomes (DSPG/sLip) are internalized by innate neutrophils, we adopted a BALB/c mouse model and assessed the neutrophil uptake of liposomes. We chose this mouse model because it is well established for neutrophil harvesting and characterization (59, 60). These experiments cannot be performed in chinchillas because many key antibodies that are required to label neutrophils are currently not available for chinchillas.

Opsonization of the hydroxylated liposomes by the complement cascade was illustrated *ex vivo* by incubating the two types of liposomes with BALB/c mouse serum for 1 hour at 37°C, followed by Western blotting to analyze their protein coronas (Fig. 1B, bottom). On the hydroxylated liposomes, DSPG/sLip, several iC3b fragments were identified: α' -67 and α' -40 (whereas α' -46 is an intermediate fragment that is not consistently captured; fig. S3C). In contrast, the protein coronas of DOTAP/sLip did not contain those complement protein fragments. The amounts of the C3b fragments displayed on the surfaces of DSPG/sLip were 40-fold higher than those on DOTAP/sLip ($P < 0.0001$).

To illustrate neutrophil liposome internalization, we isolated fresh neutrophils from mouse bone marrow and incubated them for 1 and 4 hours with opsonized liposomes labeled with 3,3'-diiodo-1,6-dioxane-2-carboxyanine (DiO). Using flow cytometry, the harvested neutrophils were gated at CD11b+/LY6G+ to confirm the cell type (61) (Fig. 1C and fig. S6A), and the fraction of neutrophils (CD11b+/LY6G+) that contains liposomes (DiO) was quantified for each type of liposome (Fig. 1C). Neutrophils incubated with phosphate-buffered saline (PBS) without liposomes were used as a control group, which showed DiO signals below the threshold (10^3). At the end of the 1-hour incubation, $86.0 \pm 2.0\%$ neutrophils were detected containing DSPG/sLip, as shown by the fraction of the incubated neutrophils that exhibited DiO signal above 10^3 , whereas for DOTAP/sLip, merely $25.0 \pm 6.3\%$ of the incubated neutrophils internalized the liposomes (Fig. 1C). Those internalization fractions remained similar after 4 hours of incubation, reaching $88.1 \pm 2.6\%$ for DSPG/sLip and $20.5 \pm 21.5\%$ for DOTAP/sLip, respectively. This result indicates that DSPG/sLip led to distinctly high internalization by neutrophils both at 1 hour ($P = 0.0003$) and 4 hours ($P = 0.0002$) compared with DOTAP/sLip.

To assess the *in vivo* neutrophil internalization efficiency, mice were injected intravenously with the two types of liposomes (DiO labeled), and flow cytometry was performed on peripheral blood (with erythrocytes removed) at 1 and 4 hours postinjection (Fig. 1D). Again, we confirmed the cell type (61) (CD11b+/LY6G+) (Fig. 1D and fig. S6B) and calculated the fraction of neutrophils (CD11b+/LY6G+) that contains liposomes (DiO) for each type of liposome (Fig. 1D). Neutrophils harvested from the peripheral blood of mice that received saline injection (without liposomes) were shown as a control group, in which the DiO signal remained below the threshold (10^3). At 1 hour postinjection, $29.6 \pm 8.2\%$ of circulating neutrophils internalized DSPG/sLip (Fig. 1D). In the *in vivo* experiment, the opsonized liposomes can be eliminated by a variety of immune mechanisms (62), thus diluting their interaction with neutrophils specifically. For DOTAP/sLip, $7.6 \pm 0.9\%$ of the sampled neutrophils internalized the liposomes. These fractions decreased slightly 4 hours postinjection because of the further clearance of the injected liposomes

by the immune system. As such, DSPG/sLip enhanced the *in vivo* neutrophil internalization to fourfold higher than that of DOTAP/sLip at 1 hour postinjection ($P = 0.0008$) and fivefold higher at 4 hours postinjection ($P = 0.0058$).

Last, to visualize the permeation of the liposome-bearing neutrophils in the TM, liposomes were labeled with 1,1'-dioctadecyl-3,3,3',3'-tetramethylindocarbocyanine perchlorate [DiI (red), a hydrophobic compound that self-assembles into the bilayer membrane of liposomes] and applied to the intact TMs of chinchillas with AOM (harvested at 72 hours postinfection). Immunocytochemistry analysis revealed that DSPG/sLip/DiI (red) overlapped with neutrophils [with elastase staining for the antigen on the neutrophil surface (green)] throughout the TM (Fig. 1E and fig. S7). The amounts of DSPG/sLip/DiI in the TM (indicated by the average DiI intensity) were 4.1-fold higher than those of DOTAP/sLip/DiI ($P = 0.0047$). To further illustrate that the liposome-bearing neutrophils can penetrate the entirety of the TM, we examined low-magnification fluorescence micrographs of both treatment groups, in which the TMs, as well as the outer and middle ear spaces, are visible (fig. S8). The overlapping green and red fluorescence throughout the TM of the group treated with DSPG/sLip/DiI hints at the hitchhiking of liposomes on neutrophils. In contrast, DOTAP/sLip/DiI predominantly accumulate in the SC connected with the external auditory canal. DSPG/sLip/DiI were also identified in the MEF, indicating that liposome-bearing neutrophils are present in the auditory bullae (Fig. 1F). The DiI intensity of DSPG/sLip/DiI was 4.1-fold higher than that of DOTAP/sLip/DiI ($P = 0.0454$) in the MEF.

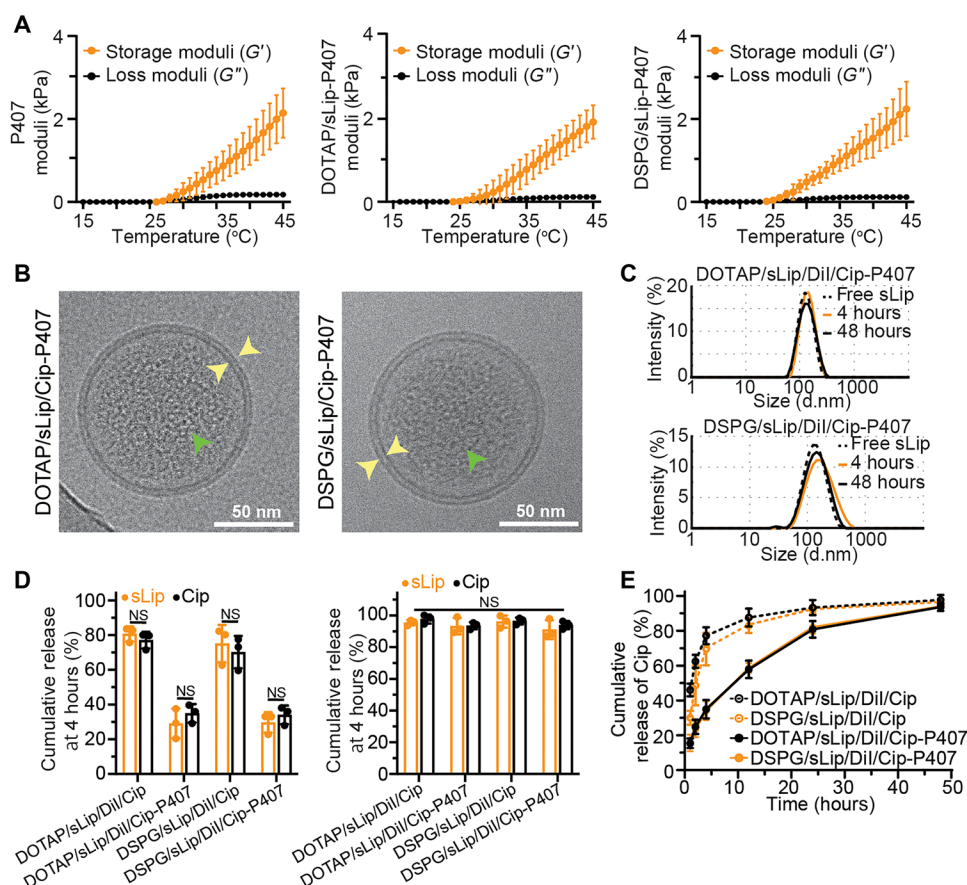
Drug-encapsulating liposomes are loaded into and released from a hydrogel intact without drug leakage

To cure AOM, it is critical that the antibiotics are delivered into the auditory bullae continuously over the course of 5 to 7 days (63, 64). To avoid premature termination of the treatment, we designed a hydrogel-based formulation to deliver the entire course of the regimen in a sustained manner from a single-dose application. We chose P407, a triblock polymer with poly(ethylene oxide) (PEO) end blocks and a poly(propylene oxide) (PPO) midblock because of desirable reverse thermal gelation at near body temperature. Increasing temperature drives the formation of spherical micelles composed of PPO cores and PEO shells (65). When a critical volume fraction of micelles is achieved, the micelles order into cubic packings, causing the low-viscosity solutions to gel (66). The hydrogel formulation thus remains a liquid during administration and gels upon contacting the warm TM (fig. S2). The ease of administration has proven key in treating pediatric patients with AOM.

We evaluated the gelation and mechanical properties of 10, 15, or 20% (w/v) solutions of P407 with or without the liposomes to identify an acceptable formulation. The 10% (w/v) P407 solution did not gel at body temperature (fig. S9A), whereas a 20% (w/v) solution had a sol-gel transition below 24°C (fig. S9B), which would cause premature gelation and poor contact with the TM. The storage (G') and loss (G'') moduli of 15% P407 were below 100 Pa at room temperature (measured by linear oscillatory shear rheology at 100 rad^{-1} , 1% strain, and 1°C min^{-1}), which increased to 994.5 ± 370.3 Pa and 167.1 ± 23.9 Pa, respectively, at body temperature (37°C) with a gelation temperature of 27° to 28°C (Fig. 2A). The P407 concentration of 15% (w/v) was therefore selected for all hydrogel formulations.

The Cip-encapsulating liposomes were loaded into a P407 hydrogel by simple mixing (67). Although amphiphiles are known to

Fig. 2. Stability, gelation, and in vitro release profiles of the liposome-containing hydrogel formulations. (A) Rheology of the hydrogel formulations, including P407, DOTAP/sLip-P407, and DSPG/sLip-P407 ($n = 3$). (B) Cryo-EM images of DOTAP/sLip/Cip-P407 and DSPG/sLip/Cip-P407. The yellow arrows indicate the lipid bilayer membranes, and the green arrows indicate the Cip crystals. Scale bars, 50 nm. (C) Size distributions of the liposomes released from the formulations tested in (E), at 4 and 48 hours; free liposomes (dotted lines) are free of P407 ($n = 3$). (D) Release of intact liposomes (sLip, detected with DiI fluorescence intensity, orange dots) and Cip (detected with HPLC, black dots) from the formulations tested in (E), at 4 and 48 hours ($n = 3$); dots represent individual samples. NS, not significant. (E) Cumulative release of Cip under infinite sink conditions over 48 hours ($n = 3$). For all panels, data are means $\pm$ SD; P values were determined by one-way ANOVA with Tukey's test; $P > 0.05$. All formulations contain 0.5 mg of Cip; in all formulations containing P407, its concentration is 15% (w/v). See fig. S9 for the rheology of formulations containing 10% (w/v) or 20% (w/v) P407.



disrupt the gelation of P407 (68, 69), the addition of liposomes did not considerably alter the rheological behavior of 15% P407 (Fig. 2A). We attributed the unchanged gelation to the presence of PEG chains distributed across the surfaces of the liposomes (5% molar ratio in the liposomes, which has been estimated to be equivalent to 50 Å of thickness) (70). The storage and loss moduli remained at 1221.8 ± 342.6 Pa and 113.1 ± 18.5 Pa for DSPG/sLip-P407 and 1023.6 ± 332.1 Pa and 98.2 ± 34.7 Pa for DOTAP/sLip-P407, respectively, at body temperature, with a gelation temperature of 27° to 28°C.

Using cryo-electron microscopy (cryo-EM), we confirmed that DSPG/sLip and DOTAP/sLip remain intact in the hydrogel after 24 hours of storage at room temperature (Fig. 2B). Their lipid bilayer membranes and the encapsulated Cip (in its crystal form because of dehydration) are identifiable under cryo-EM. The diameters of the liposomes were measured to be 112.1 ± 13.2 nm and 116.5 ± 15.1 nm for DSPG/sLip and DOTAP/sLip, respectively. These values are smaller than the diameters obtained using DLS, likely because of the shrinkage caused by the liquid nitrogen freezing as part of the cryo-EM sample preparation. The lipid membranes exhibited thicknesses of 5.0 ± 0.7 nm and 5.1 ± 0.8 nm for DSPG/sLip and DOTAP/sLip, respectively, and no rupture of the lipid membranes was observed.

The in vitro release kinetics were examined by placing the liposome-containing hydrogel in a Transwell insert and collecting the solution in the receiving chamber at 4 and 48 hours. The structural integrity of the liposomes upon release was analyzed with DLS (Fig. 2C). At 4 hours, particles in the receiving solution (0.9% NaCl solution) demonstrate average diameters of 134.1 ± 9.5 nm for DSPG/sLip/

DiI/Cip-P407 and 136.0 ± 5.0 nm for DOTAP/sLip/DiI/Cip-P407, which became 131.2 ± 5.1 nm and 140.2 ± 27.4 nm, respectively, at 48 hours. The average particle sizes are close to those of the free liposomes (dotted line in Fig. 2C), 130.7 ± 4.7 nm for DSPG/sLip/DiI/Cip and 132.5 ± 4.3 nm for DOTAP/sLip/DiI/Cip (table S1), indicating that the liposomes remain intact during the release from the P407 hydrogel.

The in vitro release kinetics of liposomes were obtained by measuring the fluorescence intensity (from DiI) of the receiving chamber solution (orange columns in Fig. 2D). Note that, because free DiI has low solubility in saline, its fluorescence is predominantly associated with intact liposomes. Upon liposome disintegration, the lipids are released into the solution, allowing Cip to rapidly diffuse. In contrast, DiI is likely to aggregate, which can diminish or entirely quench its fluorescence. Alternatively, DiI may remain associated with lipid fragments, yielding a weaker fluorescence signal compared with that from the densely packed DiI molecules in the intact liposome bilayer. At 4 hours, $29.6 \pm 5.9\%$ and $29.1 \pm 8.6\%$ of the liposomes in DSPG/sLip/DiI/Cip-P407 and DOTAP/sLip/DiI/Cip-P407 were released, respectively, which increased to $91.1 \pm 6.0\%$ and $93.3 \pm 5.1\%$ at 48 hours. The surface charge or chemistry of the liposomes did not affect their release kinetics from the P407 hydrogel. The in vitro release kinetics of Cip were analyzed using high-performance liquid chromatography (HPLC) (Fig. 2E). The results closely match those of the liposomes (Fig. 2D), indicating undetectable leakage of the encapsulated drug from the liposomes during their diffusion through the hydrogel matrix. The hydrogel slows the

overall release kinetics of Cip for the first 4 hours, from $70.2 \pm 9.3\%$ to $34.0 \pm 5.4\%$ for DSPG/sLip/Dil/Cip and from $77.1 \pm 4.9\%$ to $34.9 \pm 5.3\%$ for DOTAP/sLip/Dil/Cip, but not by 48 hours, when $94.0 \pm 2.5\%$ and $93.6 \pm 2.2\%$ Cip is released from DSPG/sLip/Dil/Cip-P407 and DOTAP/sLip/Dil/Cip-P407, respectively.

Beyond Cip, the DSPG and DOTAP liposomes are also compatible with other antibiotics, as confirmed using a variety of fluoroquinolones, including moxifloxacin hydrochloride, levofloxacin hydrochloride, and Cip base (table S2). The latter is poorly soluble in water. The liposomes exhibit comparable size, size distribution, and zeta potential regardless of the antibiotics encapsulated, hinting at similar interactions with the innate neutrophils. Therefore, their comparative treatment outcomes should be consistent with those reported in the literature (71–73). It is worth noting that Cip base leads to the greatest drug loading, which is attributed to its preferential partitioning into the lipid bilayers.

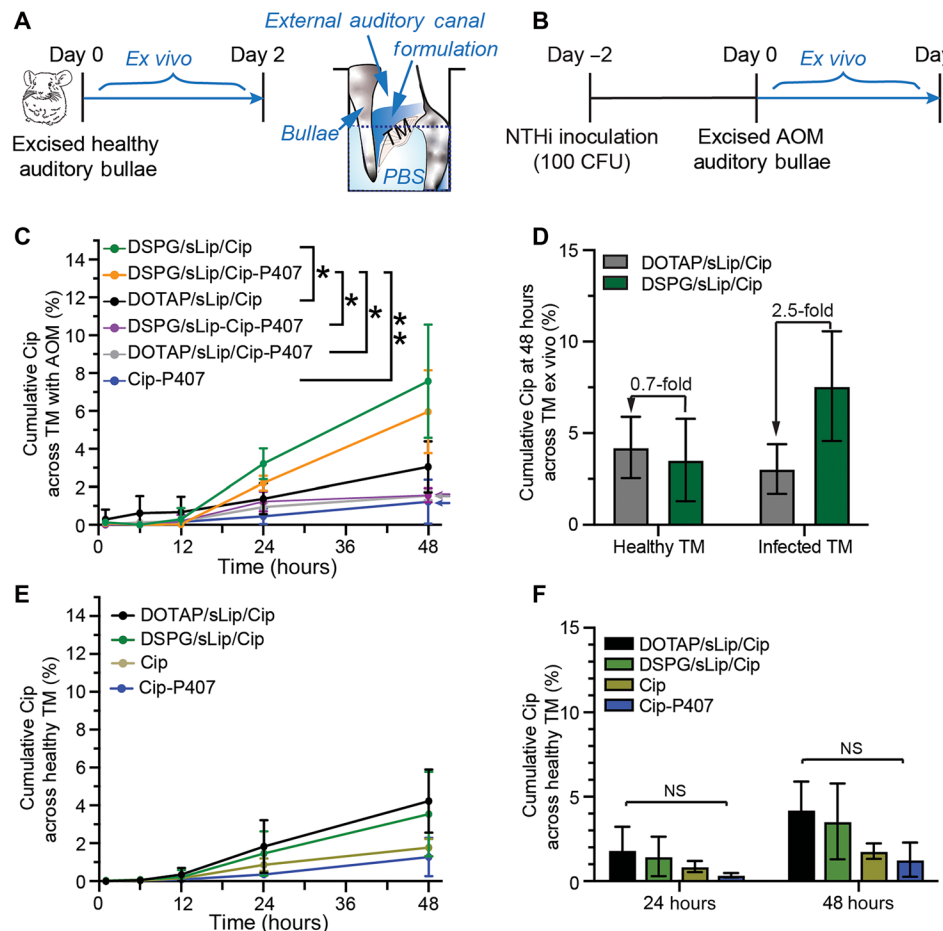
The liposome-containing hydrogel enables transtympanic antibiotic delivery ex vivo

The effectiveness of the transtympanic drug delivery strategy was demonstrated by quantifying the drug flux across the intact TM in health or disease using excised auditory bullae (Fig. 3, A and B). The intactness of the excised TM was confirmed using electrical impedance measurements (14, 18, 19, 74). The TM with microperforations was excluded from the reported results. Each test formulation contains 0.5 mg of Cip (at a concentration of 2.5 mg/ml).

Among the test formulations, DSPG/sLip/Cip shows the greatest permeability in an infected TM (Fig. 3C). Over 48 hours, $7.6 \pm 3.0\%$ of the Cip contained in this formulation permeated across the TM cumulatively, representing a 2.5-fold higher value than that delivered by DOTAP/sLip/Cip ($3.1 \pm 1.4\%$; $P = 0.0146$; Fig. 3D). This comparison contrasts with their transtympanic delivery efficiency in healthy ears, where DOTAP/sLip/Cip transported $4.2 \pm 1.7\%$ of the Cip contained in the formulation and DSPG/sLip/Cip $3.5 \pm 2.2\%$ (Fig. 3E). We further compared the Cip penetration across the healthy TMs for all treatment groups at 24 and 48 hours, and there was no statistically significant difference among the groups (DSPG/sLip/Cip versus Cip: $P = 0.5151$; DSPG/sLip/Cip versus DOTAP/sLip/Cip: $P = 0.9415$), with or without the liposome encapsulation (Fig. 3F). The overall transport remained in the passive diffusion range given that neutrophils are absent in the healthy TM (fig. S1A).

Although the overall kinetics in transtympanic drug delivery are dominated by the permeation resistance across the intact TM (14, 75), the inclusion of a hydrogel matrix can slow down the overall rate of transport (14, 76). Therefore, we assessed the ex vivo permeability of Cip achieved by the liposome-containing hydrogel using the intact TMs excised from chinchillas with AOM (Fig. 3C). Despite the additional transport resistance exerted by the hydrogel matrix, DSPG/sLip/Cip-P407 demonstrated the second highest transtympanic delivery rate in infected ears among all formulations tested. Cumulatively over 48 hours, $6.0 \pm 2.2\%$ of the Cip permeated across the TM from DSPG/sLip/Cip-P407, which is 3.8-fold higher than

Fig. 3. Ex vivo cumulative Cip penetration across the TM into a receiving chamber. (A) Diagram of the ex vivo protocol for assessing Cip permeation across the healthy TM. (B) Timeline of the ex vivo protocol in an AOM animal model; a CFU value of 100 indicates the amount of NTHi inoculated 48 hours before collecting the auditory bullae. (C) Cumulative permeation (%) of Cip across the TMs of chinchillas with AOM over 48 hours ($n = 4$). DSPG/sLip-Cip-P407 (purple line) indicates a mixture of DSPG/sLip, free Cip, and P407; $*P = 0.0146$, $*P = 0.0177$, $*P = 0.0170$, and $**P = 0.0097$ (from left to right), calculated using one-way ANOVA with Tukey's test; $*P < 0.05$; $**P < 0.01$. (D) Comparison of the permeation (%) of Cip across the TMs of healthy chinchillas in (E) and chinchillas with AOM in (C) after 48 hours of treatment using DSPG/sLip/Cip and DOTAP/sLip/Cip. (E) Cumulative permeation (%) of Cip across the TMs of healthy chinchillas over 48 hours ($n = 3$). (F) Comparison of the cumulative permeation (%) of Cip across the TMs of healthy chinchillas at 24 and 48 hours in (E); P values were determined by one-way ANOVA with Tukey's test; $P > 0.05$. Data are presented as means $\pm$ SD, and dots represent individual participants. All formulations contain 0.5 mg of Cip; where included, P407 is 15% (w/v).



that achieved by DOTAP/sLip/Cip-P407 ($P = 0.0170$) and 5-fold higher than that for Cip-P407 ($P = 0.0097$). This result also corroborates the intactness of the excised TMs, proving that therapeutics did not simply leak through the TMs by perforations. Had perforations occurred, Cip should diffuse faster than liposomes, leading to the greatest flux for unencapsulated Cip, that is, the Cip-P407 formulation. That was not the case. Cip-P407 (Fig. 3C, blue) achieved the lowest flux, and mixing (instead of encapsulating) Cip with DSPG/sLip led to no permeation enhancement effect. We formulated a hydrogel dose where Cip was mixed with DSPG/sLip, which is denoted as DSPG/sLip-Cip-P407, and quantified the cumulative transtympanic permeation across the infected TMs (hence with neutrophil presence). The cumulative Cip was $1.6 \pm 0.4\%$ ($P = 0.0177$ versus DSPG/sLip/Cip-P407) in 48 hours (Fig. 3C, purple). This result confirms that the DSPG liposomes do not function as a simple CPE and that DSPG/sLip cross the TM much faster than Cip, suggesting a possibility of active transport instead of passive diffusion. Although the ex vivo permeability measurement confirmed the viability of leveraging the innate immune response to enhance the transtympanic permeation of therapeutics, it was performed on isolated auditory bullae. The immune system (besides the chemotaxis signals that may have leaked into the receiving chamber), including the key immune cells and complement proteins, is missing or deactivated over the course of the experiment. Therefore, we used an in vivo AOM disease model to assess the overall delivery efficiency, as detailed below.

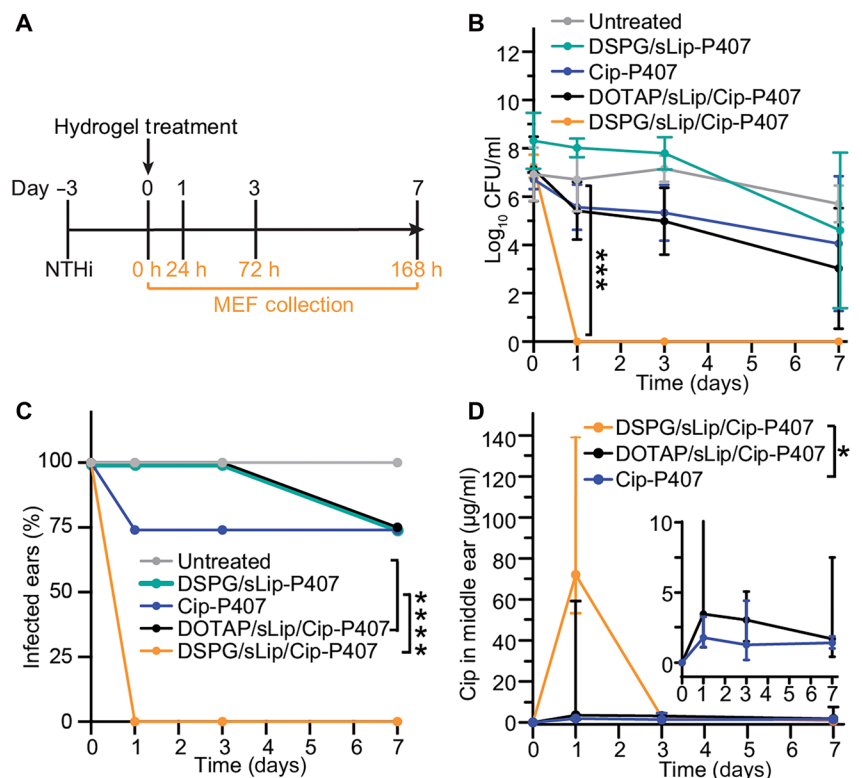
The liposome-containing hydrogel effectively treats AOM in chinchillas in vivo

Before performing the in vivo efficacy testing, we assessed the cytotoxicity of the formulations and their constituents in human dermal

fibroblasts (hFBs) (fig. S10, A and B) and pheochromocytoma cells (PC12, frequently used to test neurotoxicity) (fig. S10, C and D). We have experimented with ear-derived cell lines, such as the human nasopharyngeal epithelial cell line (Detroit 562) and human middle ear epithelial cells (HMEE) (14), but their growth is not as robust and results are less reproducible compared with the ones included in the report. We chose hFBs and PC12 for the cytotoxicity evaluation on the basis of their broad adoption in biocompatibility studies of transtympanic drug delivery systems (5, 14, 77). Although P407 hydrogel alone shows minimal cytotoxicity, the addition of 0.5 mg Cip leads to considerable cytotoxicity on day 3, with cell viability of $45.9 \pm 5.3\%$ for hFBs and $12.1 \pm 0.2\%$ for PC-12. DSPG/sLip/Cip-P407 reduced the cytotoxicity elicited by free Cip, increasing the 3-day cell viability to $88.3 \pm 7.4\%$ and $20.3 \pm 6.8\%$ for hFBs and PC12 cells, respectively.

The in vivo efficacy testing starts with the establishment of AOM in chinchillas by direct inoculation, as described above. To confirm the infection, evaluate treatment efficacy, and quantify the in vivo pharmacokinetics, the MEF was aspirated through the auditory bullae (Fig. 4A). The criterion for a successful establishment of infection is that 3 days postinoculation, the concentration in the MEF should be $>10^6$ colony-forming unit (CFU)/ml. In infected animals that did not receive any treatment, the bacterial concentration of $>10^5$ CFU/ml was maintained throughout the 7-day course of monitoring (Fig. 4B and table S3). In treated animals, 200 μ l of test formulation, which contained 0.5 mg Cip, was deposited onto each TM through the external auditory canal. The TMs were closely monitored using an otomicroscope throughout the in vivo experiment, following a previously reported procedure (18). Animals with the perforated TMs, identified by an otomicroscope or during the dissection at the end of each experiment, were excluded from the reported results.

Fig. 4. In vivo efficacy and pharmacokinetics of DSPG/sLip/Cip-P407. (A) Diagram of the in vivo experimental procedure. h, hours. (B) Time courses of bacterial CFU from the MEFs aspirated from chinchillas with AOM treated with DSPG/sLip-P407, Cip-P407, DOTAP/sLip/Cip-P407, and DSPG/sLip/Cip-P407 ($n = 4$). (Log₁₀ CFU is set to 0 instead of minus infinity for the purpose of this illustration.) Data are presented as means $\pm$ SD, and dots represent individual participants. *** $P = 0.0001$ by one-way ANOVA with Tukey's test; *** $P < 0.001$. (C) Percentages of animals with AOM from NTHi before and after receiving DSPG/sLip-P407, Cip-P407, DOTAP/sLip/Cip-P407, and DSPG/sLip/Cip-P407. Day 0 reflects the status immediately before the administration of therapeutics ($n = 4$). **** $P < 0.0001$ by Fisher's exact test. (D) Concentrations of Cip over time in the MEFs of the same animals as in (B) and (C) ($n = 4$). Inset is the magnified drug concentration range of 0 to 10 μ g/ml. Data are medians with the 25th and 75th percentiles in parentheses. * $P = 0.0142$ by the non-parametric Kruskal-Wallis multiple comparisons test; * $P < 0.05$. Each formulation contains 0.5 mg of Cip, 2 mg of lipids, and 15% (w/v) P407, unless otherwise specified. See table S3 for the normality test of data in (B) and (D) with the Shapiro-Wilk test.



In infected animals that received DOTAP/sLip/Cip-P407 or Cip-P407, 25% of treated ears were no longer infected with NTHi by day 7 ($n = 4$, Fig. 4C). The titers of pathogens in the MEFs remained high, at $10^{3.0} \pm 10^{2.5}$ CFU/ml for DOTAP/sLip/Cip-P407 ($P = 0.2482$ versus untreated) and $10^{4.1} \pm 10^{2.8}$ CFU/ml for Cip-P407 ($P = 0.6270$ versus untreated) (Fig. 4B). An antibiotic-free formulation, DSPG/sLip-P407, was also tested to assess whether the innate immune response alone can clear AOM. This formulation achieved a 25% cure rate, with an MEF pathogen titer of $10^{4.6} \pm 10^{3.3}$ CFU/ml ($P = 0.9563$ versus untreated). In contrast, 100% of the animals receiving DSPG/sLip/Cip-P407 did not have infected ears within the first 24 hours of the treatment, as indicated by the zero CFU in the MEFs (Fig. 4B) ($P = 0.0001$ versus untreated). The sterile auditory bullae were maintained until the end of the experiment on day 7.

The 100% cure rate was a direct result of the large fraction of Cip delivered into the middle ear by DSPG/sLip/Cip-P407 (Fig. 4D). As quantified using HPLC, a peak Cip concentration of 72.0 $\mu\text{g/ml}$ [median; interquartile range (IQR) = 53.3 to 139.1 $\mu\text{g/ml}$] was achieved by DSPG/sLip/Cip-P407, representing an up to 600-fold higher value than the reported minimum inhibitory concentration (MIC) of Cip against NTHi (78). Assuming a middle ear volume of 1 ml for a chinchilla (79), 14.4% (median; IQR = 10.7% to 27.8%) of the total Cip contained in DSPG/sLip/Cip-P407 was delivered across the TM within 24 hours (fig. S11). That delivery efficiency (at day 1) was 21-fold higher than that achieved by DOTAP/sLip/Cip-P407, that is, 0.7% (peak median concentration = 3.5 $\mu\text{g/ml}$; IQR = 1.1 to 59.2 $\mu\text{g/ml}$; $P = 0.0955$ versus DSPG/sLip/Cip-P407), and 36-fold higher than that achieved by Cip-P407, that is, 0.4% (peak median concentration = 1.8 $\mu\text{g/ml}$; IQR = 1.1 to 3.3 $\mu\text{g/ml}$; $P = 0.0142$ versus DSPG/sLip/Cip-P407).

The delivery efficiency of DSPG/sLip/Cip-P407 can be further increased upon exacerbation of the native immune response. We used a coinfection animal model (where animals were simultaneously infected with NTHi and *Streptococcus pneumoniae*) to achieve that heightened immune response. The treatment with 200 μl of DSPG/sLip/Cip-P407 led to a median Cip concentration of 262.4 ± 211.7 $\mu\text{g/ml}$ in the MEF after 24 hours, representing a median delivery efficiency of $52.5 \pm 42.3\%$ (fig. S12). The experiment was terminated on day 1 because of the development of pneumonia and high rates of unanticipated sudden death even with close monitoring. Given that no perforations were observed in the TMs of the treated animals, we believe the higher delivery efficiency was a result of the heightened immune reactions. This can be further elucidated by comparing the 24-hour cumulative Cip levels in ex vivo and in vivo models, where a 20.9-fold enhancement by DSPG/sLip was observed in vivo, but only a 2.2-fold enhancement was observed ex vivo (fig. S13). A direct connection between the transtympanic delivery efficiency and the complement cascade is detailed in the section below.

The complement cascade is key to in vivo transtympanic drug delivery and successful AOM treatment

To further illustrate the key role played by the complement cascade in vivo, we depleted the complement protein fragments from the systemic circulation of the infected animals and demonstrated that the transtympanic delivery was no longer effective. The complement proteins were depleted by treating infected animals with cobra venom factor (CVF), a structural analog of C3, and a nontoxic protein purified from cobra venom (80). The treatment with CVF has been reported to deplete complement proteins and achieve an anti-complement

effect by its continuous and sustained activation of C3 (80, 81). CVF was injected at a dose of 500 mg/kg body weight (through the femoral vein) 24 hours before the hydrogel treatment (Fig. 5A). The C3b and its iC3b fragments in the MEFs and sera of the treated animals were undetectable by Western blotting on day 1 and remained nearly undetectable for up to 3 days posthydrogel treatment (Fig. 5B).

In the animals that received CVF, DSPG/sLip/Cip-P407 ameliorated NTHi in 25% of the infected ears by day 7 (Fig. 5C), with an NTHi titer of $10^{4.0} \pm 10^{2.7}$ CFU/ml in the MEFs on day 7 (Fig. 5D). The results for animals treated with DSPG/sLip/Cip-P407 and no CVF (Fig. 4) are replotted in Fig. 5 to serve as a point of comparison. The Cip concentration in the MEFs of the CVF-treated animals was 2.7 ± 3.2 $\mu\text{g/ml}$, captured on day 1, corresponding to $0.5 \pm 0.6\%$ of the total Cip in the formulation (Fig. 5E). As such, there was a 33-fold difference in MEF drug concentration in CVF-treated animals versus nontreated ones ($P = 0.0124$; Fig. 5F).

CVF is also known to deplete C5a (82), a complement component and a neutrophil chemoattractant released from tissues near the site of infection (83). C5a plays a critical role in the local recruitment of neutrophils given that it is known to override the signals by chemoattractants released by intermediary sites, such as the endothelium (84), which has been considered a dominating mechanism to trigger neutrophil chemotaxis (85). Therefore, the treatment with CVF likely also reduces the local recruitment of neutrophils, which further contributes to the reduced efficacy. Together, the results from the CVF treatment demonstrate that the complement cascade is essential for achieving therapeutic antibiotic concentrations in the middle ear.

The liposome-containing hydrogel is biocompatible in the ear

The TMs harvested from infected but untreated animals exhibited an average thickness of 57.4 ± 11.2 μm , 5.8-fold greater than that of the pristine TM (9.9 ± 2.0 μm) (Fig. 6A). Acute inflammation was observed with diffuse edema and dense infiltration by inflammatory cells. The TMs excised from the animals infected and treated with DSPG/sLip/Cip-P407 for 7 days appeared nearly identical to the pristine ones, with an average thickness of 12.3 ± 1.4 μm and no tissue injuries, necrosis, or inflammatory cells (Fig. 6A). By comparison, animals that received Cip-P407 (40.3 ± 12.0 μm), DOTAP/sLip/Cip-P407 (46.0 ± 7.4 μm), DSPG/sLip-P407 (58.5 ± 13.0 μm), or CVF injection before the DSPG/sLip/Cip-P407 treatment (50.1 ± 12.1 μm) demonstrated considerable edema and inflammation in their TMs (Fig. 6A). The mucosal layer that lines the middle ear bullae exhibited an appearance similar to that of the pristine bullae after the treatment with DSPG/sLip/Cip-P407 for 7 days (fig. S14), consistent with good local tolerability of the treatment in the ear.

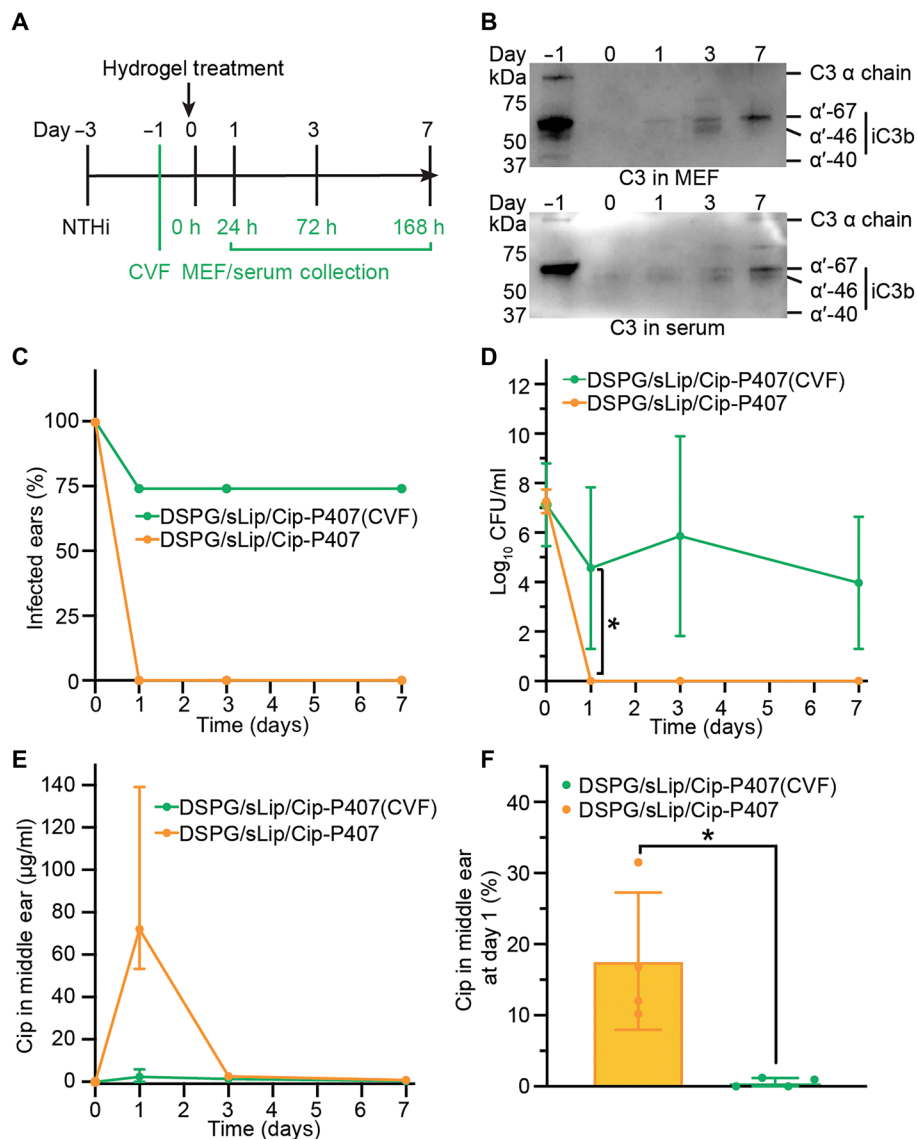
In addition, the biocompatibility of DSPG/sLip/Cip-P407 in the ear was evaluated on the basis of its effects on inner ear hair cells and hearing sensitivity. As shown in Fig. 6B, the morphology and arrangement of the three layers of outer hair cells (OHCs) and one layer of inner hair cells (IHCs) are similar to the pristine ones (Fig. 6B-i) after 7 days of treatment with the DSPG/sLip/Cip-P407 formulation (Fig. 6B-ii). The effect of DSPG/sLip/Cip-P407 on hearing sensitivity was assessed by the auditory brainstem response (ABR) of healthy chinchillas before and after receiving the formulation. Placement of 200 μl of DSPG/sLip/Cip-P407 on the TM did not cause a detectable shift in the ABR threshold (fig. S15), indicating unchanged hearing sensitivity as detailed in a previous study (14).

Fig. 5. In vivo efficacy of DSPG/sLip/Cip-P407 in C3b-depleted animals. (A) Diagram of the in vivo experimental procedure to deplete C3b fragments, using CVF, in infected chinchillas. (B) Western blot showing the amounts of C3b fragments over time in the MEFs and sera sampled from the CVF-treated chinchillas [same in (C) to (F)]. (C) Percentages of animals with AOM after receiving DSPG/sLip/Cip-P407, with and without the CVF treatment ($n = 4$). (D) Time course of the bacterial CFU from the MEFs of the same animals as in (B) and (C) ($n = 4$). * $P = 0.0313$ by two-tailed unpaired Student's t test. (E) Concentrations of Cip over time in the MEFs of the same animals as in (B) and (D) ($n = 4$). (F) Cumulative permeation of Cip across the TMs in the first 24 hours of the hydrogel treatment (in terms of a percentage of the total Cip applied) ($n = 4$). * $P = 0.0124$ by two-tailed unpaired Student's t test. The data on the animals that received DSPG/sLip/Cip-P407 and no CVF are replotted from Fig. 4 as a point of comparison. Data are means $\pm$ SD; dots represent individual participants; * $P < 0.05$. See fig. S11 for the corresponding cumulative transtympanic permeation (%) of Cip over time.

DISCUSSION

We have developed a fresh concept in transtympanic drug delivery, which led to a formulation that cures AOM by targeting antibiotics directly to the middle ear. The liposome-based drug delivery system achieves a transtympanic antibiotic delivery efficiency of 14%, corresponding to a four- to eightfold higher value than that for the best-in-class formulations that use CPEs to disrupt the TM (14, 15, 19). The formulation, when placed in the outer ear canal of chinchillas infected by NTHi, eradicates AOM in 100% of the treated animals within 24 hours, using 0.5 mg of antibiotic. Furthermore, it was designed to allow a pediatrician to deliver a single dose of antibiotics to the TM that provides an entire course of treatment, reducing pediatric antibiotic usage and improving patient compliance.

The TM is a trilayer membrane, where the outermost layer is a keratinized stratified squamous epithelium continuous with the skin of the external auditory canal (86). It is one of the most impermeable biological barriers, distinguishing it from other tissues in the ear, including the middle ear mucosa, round window membrane, and tissues that do not contain stratified squamous epithelial cells (86). Like in the skin, the barrier function of the TM predominantly resides in the SC layer (51). The transtympanic delivery of therapeutics has been a long-standing challenge and the focus of many recent studies on AOM treatment (14, 19, 87–89). Although prior studies have explored transtympanic delivery using various nanocarriers, including transferosomes (90), cationic nanoparticles (91), nanowires (17), and lecithin-based liposomes (92, 93), they have not cleared AOM in vivo. The current state-of-the-art approaches often rely on the structural disruption of the TM, for example, by tympanostomy or molecular disruption using CPEs. The reported strategy is conceptually distinct given that it leverages the innate immune response of children with AOM to overcome the impermeable TM.



Leveraging the innate neutrophils as drug carriers also leads to a targeted therapeutic effect because neutrophils migrate to the site of infection in response to invading pathogens. Although neutrophils are known to infiltrate the middle ear mucosa (53, 94), their ability to penetrate the TM is unexpected because the TM is known to be impermeable to even small molecules, like helium gas (95) (with a molecular diameter of 2.6 Å). The paradigm of hitchhiking on immune cells to overcome the SC could be broadly applicable to treating diseases beyond AOM, such as topical and on-demand treatments for skin disorders. External biological barriers, such as bacterial biofilms that often cause chronic or untreatable infections, are also potential targets for the penetration effects by immune cell hitchhiking (96, 97).

The design also overcomes a challenge in harnessing neutrophils as drug delivery vehicles, that is, the short lifetime of neutrophils ($t_{1/2} < 24$ hours) (98, 99). Given that most existing systems rely on harvesting neutrophils ex vivo for drug loading, their short lifetime, combined with the laborious procedures of neutrophil isolation, purification, drug loading, and reinfusion into the bloodstream, limits

Figure 1 consists of eight histological images and a bar graph. The images show the thickness of the tympanic membrane (TM) in different experimental groups. The bar graph quantifies the TM thickness in micrometers (μm) for each group.

Histological Images:

- Pristine TM:** Shows a thin, normal TM.
- TM with AOM; untreated:** Shows a significantly thickened TM.
- TM with AOM; treated with Cip-P407:** Shows a TM that is thinner than the untreated AOM group.
- TM with AOM; treated with DOTAP/sLip/Cip-P407:** Shows a TM that is thinner than the untreated AOM group.
- TM with AOM; treated with DSPG/sLip-P407:** Shows a TM that is thinner than the untreated AOM group.
- TM with AOM; treated with DSPG/sLip/Cip-P407 (CVF):** Shows a TM that is thinner than the untreated AOM group.
- TM with AOM; treated with DSPG/sLip/Cip-P407:** Shows a TM that is thinner than the untreated AOM group.

Bar Graph:

The bar graph shows the thickness of the TM (μm) for each group. The y-axis ranges from 0 to 80 μm. The x-axis lists the groups: Pristine TM, TM with AOM (Untreated), TM with AOM treated with Cip-P407, TM with AOM treated with DOTAP/sLip/Cip-P407, TM with AOM treated with DSPG/sLip-P407, TM with AOM treated with DSPG/sLip/Cip-P407 (CVF), and TM with AOM treated with DSPG/sLip/Cip-P407. A legend indicates that hatched bars represent 'TM with AOM' and white bars represent 'NS' (Normal Saline). A bracket with three asterisks (***) indicates a significant difference between the untreated AOM group and the treated groups.

Group	Thickness of TM (μm)
Pristine TM	~10
TM with AOM (Untreated)	~55
TM with AOM treated with Cip-P407	~40
TM with AOM treated with DOTAP/sLip/Cip-P407	~45
TM with AOM treated with DSPG/sLip-P407	~55
TM with AOM treated with DSPG/sLip/Cip-P407 (CVF)	~48
TM with AOM treated with DSPG/sLip/Cip-P407	~10

leading to low delivery efficiency (101). We overcome these challenges by delivering the neutrophil-hitchhiking liposomes locally and topically, escaping systemic clearance while avoiding off-target effects. Neutrophil counts throughout an episode of AOM are a

critical consideration during the bench-to-bedside translation of this technology. Recent clinical studies indicate that the neutrophil counts in the ear, nose, and throat (ENT) systems are consistently two to four orders of magnitude greater in children with AOM than in healthy children (102). Although these studies lend strong support to the potential of the reported drug delivery system in addressing pediatric AOM, we emphasize that further investigations (for example, on the intervention timing) are required before expanding this strategy for clinical treatment.

To load antibiotics into innate neutrophils, we designed hydroxylated liposomes that are preferentially opsonized by the C3b fragments for enhanced neutrophil internalization. The preferential opsonization was illustrated using Western blotting. The effective hitchhiking of hydroxylated liposomes on neutrophils was demonstrated in a mouse model using flow cytometry. The traversing of the liposome-bearing neutrophils across the intact chinchilla TM was visualized using immunocytochemistry analysis. To further illustrate the critical role played by the complement system, we treated the infected chinchillas with CVF, which has been shown to deplete complement proteins (80, 81), before receiving the liposomes. Merely 25% of the animals cleared their AOM, in contrast to the 100% cure rate in the ones not treated with CVF, which demonstrated that the hydroxylated liposomes no longer enhance transtympanic permeation or eradicate AOM. The mammalian SC comprises broad lamellar sheets of negatively charged lipids, including ceramides (40%), free fatty acids (25%), cholesterol (25%), and cholesteryl sulfate (10%) (103). Therefore, positively charged liposomes, such as DOTAP/sLip, have been considered favorable for enhancing transtympanic permeation by electrostatic interactions (104, 105), which is consistent with our ex vivo results on the permeability of the healthy TM. However, during AOM, the hydroxylated liposome DSPG/sLip exhibit a superior transtympanic permeability than the positively charged liposomes, reversing the trend observed in healthy ears. The in vivo transtympanic delivery efficiency achieved by DSPG/sLip was sevenfold higher than its ex vivo efficiency after 24 hours of treatment, which is the opposite of what has been observed in previous transtympanic delivery studies (14, 19, 87, 88). That greater delivery efficiency in vivo hints at the enabling role played by the innate immune system, which is only present in live animals. Furthermore, we unveiled a positive correlation between the transtympanic delivery efficiency and the severity of the infection, pointing to a smart and self-regulated treatment.

The formulation comprises a temperature-responsive hydrogel that is easy to apply under room temperature and gels quickly upon contacting the warm TM, streamlining the treatment of young patients who are most prone to AOM. A single dose can be applied by a pediatrician at the time of diagnosis, which delivers the entire course of the regimen to cure the infection. P407 was used to formulate the temperature-responsive hydrogel. Using cryo-EM, rheology, and DLS, we confirmed that the liposomes remain intact upon loading into or release from the P407 hydrogel, and the liposomes do not disrupt the reverse thermal gelation of P407. Furthermore, the reverse thermal gelation of P407-based hydrogels has been validated in humans, for example, through administration by a tympanostomy tube into the middle ear for the sustained release of antibiotics (106–108). The formulation deposition onto the TM can be done using an otoscope, which is anticipated to be more successful in human ears than in chinchilla ears because of the straight human ear canal. In the unlikely event that gelation occurs on the wall of the external auditory canal, it can be easily removed by a water rinse,

thus unlikely to affect the ultimate therapeutic efficacy. Thus, P407, which has been approved by the FDA as a polymeric drug carrier, combined with the many FDA-approved liposomal formulations, promises rapid progression of the reported treatment toward clinical adoption.

Cip was chosen on the basis of its broad-spectrum activity against relevant bacteria and its effectiveness against AOM in children with myringotomy tubes, and liposomes are known for their encapsulation versatility. Although the FDA has warned against the use of systemic quinolones because of the increased risk of tendinitis and tendon rupture (109–114), such systemic toxicity is not a main concern with topical therapy (115). Locally applied quinolones, especially Cip, remain a common option to treat patients with a perforated TM (116), which has informed our choice of Cip as the local therapeutic in this study. Furthermore, the delivery system is likely compatible with other drugs. We have confirmed that the reported liposomes can be used to deliver a variety of antibiotics, including those with poor bioavailability and solubility, which could be leveraged to broaden the range of antimicrobials applicable in AOM treatment.

The limitations of this work are detailed below. The correlation between particle size and neutrophil uptake efficiency, the involvement of other immune cells in transtympanic delivery (especially in chronic/recurrent OM), potential barriers as we move toward a human clinical trial, and additional preclinical studies that may be required are not well understood. Although liposomes with a diameter of around 100 nm were used in this study, on the basis of the reported preference of neutrophils for nanometer-scale particles compared with micrometer ones (117, 118), there is room for further optimization. In addition, whereas the current study focuses on the effect of neutrophils in transtympanic delivery, it is possible that other immune cells, particularly macrophages, may serve as additional liposome delivery vehicles. Although not observed in the infected TMs and MEFs in this study (fig. S1), macrophages are known to contribute to the clearance of pathogens and cellular debris during AOM (46, 119). They produce proinflammatory cytokines that orchestrate the recruitment and activation of additional immune cells (119–121), and their activity is closely associated with the progression of persistent or recurrent OM cases (122, 123). Future work exploring this potential of macrophages will likely involve the development of chronic/recurrent OM or other long-term infections. Although *C. lanigera* is one of the most prevalent animal models for AOM studies, the safety and efficacy results of which have supported human clinical trial applications (124), we acknowledge that the TM thickness and immune responses can differ from human children. We emphasize that the strategy presented here leverages the active transport of neutrophils instead of passive diffusion (for example, for CPEs). Despite the thicker human TM that is anticipated to slow down passive diffusion, that effect is less likely to affect active transport considerably. Regarding immune responses, although direct evidence of the neutrophil infiltration in the human TM during AOM is not available in the literature (likely because of the scarcity of such TM samples), a recent report confirmed the presence of neutrophils in the middle ear purulent effusion in an 18-month-old girl with acute onset of OM (125), pointing to the translation potential of the reported technology. Nevertheless, the limited number of comparative studies of the human immune responses versus those of chinchillas could reduce the confidence in any prediction made regarding the ultimate efficacy in humans. Additional preclinical studies may be required to address the potential for antibiotic

resistance and efficacy in polymicrobial AOM episodes. The low dose of Cip administered does not exclude the possibility that antibiotic drainage into the nasopharynx could breed antibiotic resistance among the nasopharyngeal flora. In some cases, OM is a polymicrobial disease that implicates multiple bacterial and viral pathogens (126, 127). The reported formulation may demonstrate limited efficacy against viral infections, but it could be adapted in future work to deliver antiviral and antibiotic therapeutics simultaneously.

MATERIALS AND METHODS

Study design

We aimed to develop a noninvasive, transtympanic drug delivery platform for OM. We focused on developing and evaluating formulations containing liposomes and an antibiotic. The experiments compared the effect of the polymer matrix and incorporation of liposomes on TM permeability and OM cure rate. During *ex vivo* experiments, data collection was stopped after 48 hours due to microbial growth on the harvested TM, and the TM with microperforations was excluded from the reported results. During *in vivo* experiments, data collection was stopped after 7 days because OM would either be cleared or cause the animal severe illness that required euthanasia, and animals with the perforated TM at the end of experiment were excluded from the reported results. Details of composition and drug loading are available in table S4. For the *in vivo* experiments, a sample size of 4 is used, which is supported by prior publications in this field (5, 19, 88). *In vivo* experiments were blinded. All key findings were replicated across independent experiments. The information regarding whether the animal was infected or healthy and regarding which formulation was administered was kept by blinded by two separate research groups and not shared until all data analyses were done. Animals were assigned treatment groups at random.

Statistical analyses

Statistical analyses were conducted using GraphPad software (version 9.4.1, San Diego, CA). All statistical analyses used are indicated in the corresponding figure captions. The Shapiro-Wilk test was applied to test the normality of the data (128), and the Shapiro-Wilk test results are shown in table S3. Normally distributed data are described with mean and SD and analyzed using two-tailed unpaired Student's *t* tests for two-group comparisons or one-way analysis of variance (ANOVA) with Tukey's test for multiple-group comparisons. When data deviate from normal distributions (Fig. 4D), medians with the IQR in parentheses are used; statistical analyses were performed with the nonparametric Kruskal-Wallis test for multiple comparisons. Data in Fig. 4C were analyzed by Fisher's exact test. *P* < 0.05 was considered statistically significant. All individual-level data are available in data file S1.

Supplementary Materials

The PDF file includes:

Materials and Methods
Figs. S1 to S15
Tables S1 to S4
References (129–133)

Other Supplementary Material for this manuscript includes the following:

Data file S1
MDAR Reproducibility Checklist

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submitted by Cornell University that covers "Native neutrophil-mediated transtympanic drug delivery to minimize antibiotic exposure in otitis media treatment" (PCT/US2025/15435). W.T. and R.Y. are cofounders and advisors for Oto Biotech Inc., a company that develops transtympanic delivery strategies, and own equity in this company. R.Y. holds a management position in Oto Biotech. The other authors declare that they have no competing interests.

Data, code, and materials availability: All data associated with this study are present in the paper or the Supplementary Materials. Raw tabular data are available in data file S1. All materials used or generated in this study are commercially available or will be supplied upon

reasonable request to R.Y. The newly generated liposome and hydrogel materials can be shared upon reasonable request through a material transfer agreement.

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Neutrophil-mediated delivery of antibiotics across the tympanic membrane for otitis media treatment in a preclinical model

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Editor's summary

Acute otitis media is a common middle ear infection that is currently treated with systemic antibiotics. The impermeable tympanic membrane that separates the outer and middle ear is a barrier to local sustained delivery of antibiotics, which could reduce side effects and increase compliance. Here, Tang *et al.* developed a strategy for neutrophil-mediated transport of antibiotics across the tympanic membrane by liposomal targeting of neutrophils. In ex vivo and in vivo animal models, the liposomes were taken up by neutrophils and crossed the tympanic membrane. In models of otitis media, topical application of their liposomal formulation in an optimized hydrogel promoted enrichment of antibiotic in the middle ear and resolution of infection in a chinchilla model. The system offers a strategy for cell-mediated delivery of antibiotics across the tympanic membrane as a potential local therapeutic for otitis media. —Molly Ogle

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